

Chapter 9. Supplemental Text Material

S9.1. Yates's Algorithm for the 3^k Design

Computer methods are used almost exclusively for the analysis of factorial and fractional designs. However, Yates's algorithm can be modified for use in the 3^k factorial design. We will illustrate the procedure using the data in Example 5.1. The data for this example are originally given in Table 5.1. This is a 3^2 design used to investigate the effect of material type (A) and temperature (B) on the life of a battery. There are $n = 4$ replicates.

The Yates' procedure is displayed in Table 1 below. The treatment combinations are written down in standard order; that is, the factors are introduced one at a time, each level being combined successively with every set of factor levels above it in the table. (The standard order for a 3^3 design would be 000, 100, 200, 010, 110, 210, 020, 120, 220, 001, . . .). The Response column contains the total of all observations taken under the corresponding treatment combination. The entries in column (1) are computed as follows. The first third of the column consists of the sums of each of the three sets of three values in the Response column. The second third of the column is the third minus the first observation in the same set of three. This operation computes the linear component of the effect. The last third of the column is obtained by taking the sum of the first and third value minus twice the second in each set of three observations. This computes the quadratic component. For example, in column (1), the second, fifth, and eighth entries are $229 + 479 + 583 = 1291$, $-229 + 583 = 354$, and $229 - (2)(479) + 583 = -146$, respectively. Column (2) is obtained similarly from column (1). In general, k columns must be constructed.

The Effects column is determined by converting the treatment combinations at the left of the row into corresponding effects. That is, 10 represents the linear effect of A , A_L , and 11 represents the AB_{LXL} component of the AB interaction. The entries in the Divisor column are found from

$$2^r 3^t n$$

where r is the number of factors in the effect considered, t is the number of factors in the experiment minus the number of linear terms in this effect, and n is the number of replicates. For example, B_L has the divisor $2^1 \times 3^1 \times 4 = 24$.

The sums of squares are obtained by squaring the element in column (2) and dividing by the corresponding entry in the Divisor column. The Sum of Squares column now contains all of the required quantities to construct an analysis of variance table if both of the design factors A and B are quantitative. However, in this example, factor A (material type) is qualitative; thus, the linear and quadratic partitioning of A is not appropriate. Individual observations are used to compute the total sum of squares, and the error sum of squares is obtained by subtraction.

Table 1. Yates's Algorithm for the 3^2 Design in Example 5.1

Treatment Combination	Response	(1)	(2)	Effects	Divisor	Sum of Squares
00	539	1738	3799			
10	623	1291	503	A_L	$2^1 \times 3^1 \times 4$	10,542.04
20	576	770	-101	A_Q	$2^1 \times 3^2 \times 4$	141.68
01	229	37	-968	B_L	$2^1 \times 3^1 \times 4$	39,042.66
11	479	354	75	AB_{LXL}	$2^2 \times 3^0 \times 4$	351.56
21	583	112	307	AB_{QXL}	$2^2 \times 3^1 \times 4$	1,963.52
02	230	-131	-74	B_Q	$2^1 \times 3^2 \times 4$	76.96
12	198	-146	-559	AB_{LXQ}	$2^2 \times 3^1 \times 4$	6,510.02
22	342	176	337	AB_{QXQ}	$2^2 \times 3^2 \times 4$	788.67

The analysis of variance is summarized in Table 2. This is essentially the same results that were obtained by conventional analysis of variance methods in Example 5.1.

Table 2. Analysis of Variance for the 3^2 Design in Example 5.1

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P-value
$A = A_L \times A_Q$	10,683.72	2	5,341.86	7.91	0.0020
B, Temperature	39,118.72	2	19,558.36	28.97	<0.0001
B_L	(39,042.67)	(1)	39,042.67	57.82	<0.0001
B_Q	(76.05)	(1)	76.05	0.12	0.7314
AB	9,613.78	4	2,403.44	3.576	0.0186
$A \times B_L = AB_{LXL} + AB_{QXL}$	(2,315.08)	(2)	1,157.54	1.71	0.1999
$A \times B_Q = AB_{LXQ} + AB_{QXQ}$	(7,298.70)	(2)	3,649.75	5.41	0.0106
Error	18,230.75	27	675.21		
Total	77,646.97	35			

S9.2. Aliasing in Three-Level and Mixed-Level Designs

In Chapter 8 we gave a general method for finding the alias relationships for a fractional factorial design. Fortunately, there is a general method available that works satisfactorily in many situations. The method uses the polynomial or regression model representation of the model,

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$$

where $\mathbf{y}$ is an $n \times 1$ vector of the responses, $\mathbf{X}_1$ is an $n \times p_1$ matrix containing the design matrix expanded to the form of the model that the experimenter is fitting, $\boldsymbol{\beta}_1$ is an $p_1 \times 1$ vector of the model parameters, and $\boldsymbol{\varepsilon}$ is an $n \times 1$ vector of errors. The least squares estimate of $\boldsymbol{\beta}_1$ is

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$$

The **true** model is assumed to be

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$$

where $\mathbf{X}_2$ is an $n \times p_2$ matrix containing additional variables not in the fitted model and $\boldsymbol{\beta}_2$ is a $p_2 \times 1$ vector of the parameters associated with these additional variables. The parameter estimates in the fitted model are not unbiased, since

$$\begin{aligned} E(\hat{\boldsymbol{\beta}}_1) &= \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 \\ &= \boldsymbol{\beta}_1 + \mathbf{A}\boldsymbol{\beta}_2 \end{aligned}$$

The matrix $\mathbf{A} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2$ is called the **alias matrix**. The elements of this matrix identify the alias relationships for the parameters in the vector $\boldsymbol{\beta}_1$.

This procedure can be used to find the alias relationships in three-level and mixed-level designs. We now present two examples.

Example 1

Suppose that we have conducted an experiment using a 3^2 design, and that we are interested in fitting the following model:

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_{12}x_1x_2 + \beta_{11}(x_1^2 - \bar{x}_1^2) + \beta_{22}(x_2^2 - \bar{x}_2^2) + \boldsymbol{\varepsilon}$$

This is a complete quadratic polynomial. The pure second-order terms are written in a form that orthogonalizes these terms with the intercept. We will find the aliases in the parameter estimates if the true model is a reduced cubic, say

$$\begin{aligned} y &= \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_{12}x_1x_2 + \beta_{11}(x_1^2 - \bar{x}_1^2) + \beta_{22}(x_2^2 - \bar{x}_2^2) \\ &\quad + \beta_{111}x_1^3 + \beta_{222}x_2^3 + \beta_{122}x_1x_2^2 + \boldsymbol{\varepsilon} \end{aligned}$$

Now in the notation used above, the vector $\boldsymbol{\beta}_1$ and the matrix $\mathbf{X}_1$ are defined as follows:

$$\beta_1 = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_{12} \\ \beta_{11} \\ \beta_{22} \end{bmatrix}, \text{ and } \mathbf{X}_1 = \begin{bmatrix} 1 & -1 & -1 & 1 & 1/3 & 1/3 \\ 1 & -1 & 0 & 0 & 1/3 & -2/3 \\ 1 & -1 & 1 & -1 & 1/3 & 1/3 \\ 1 & 0 & -1 & 0 & -2/3 & 1/3 \\ 1 & 0 & 0 & 0 & -2/3 & -2/3 \\ 1 & 0 & 1 & 0 & -2/3 & 1/3 \\ 1 & 1 & -1 & -1 & 1/3 & 1/3 \\ 1 & 1 & 0 & 0 & 1/3 & -2/3 \\ 1 & 1 & 1 & 1 & 1/3 & 1/3 \end{bmatrix}$$

Now

$$\mathbf{X}_1' \mathbf{X}_1 = \begin{bmatrix} 9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

and the other quantities we require are

$$\mathbf{X}_2 = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 0 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \beta_2 = \begin{bmatrix} \beta_{111} \\ \beta_{222} \\ \beta_{122} \end{bmatrix}, \text{ and } \mathbf{X}_1' \mathbf{X}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 6 & 0 & 4 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The expected value of the fitted model parameters is

$$E(\hat{\beta}_1) = \beta_1 + (\mathbf{X}_1' \mathbf{X}_1)^{-1} \mathbf{X}_1' \mathbf{X}_2 \beta_2$$

or

$$E \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_{12} \\ \hat{\beta}_{11} \\ \hat{\beta}_{22} \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_{12} \\ \beta_{11} \\ \beta_{22} \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 0 & 0 & 0 \\ 6 & 0 & 4 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta_{111} \\ \beta_{222} \\ \beta_{122} \end{bmatrix}$$

The alias matrix turns out to be

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 2/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This leads to the following alias relationships:

$$\begin{aligned} E(\hat{\beta}_0) &= \beta_0 \\ E(\hat{\beta}_1) &= \beta_1 + \beta_{111} + (2/3)\beta_{122} \\ E(\hat{\beta}_2) &= \beta_2 + \beta_{222} \\ E(\hat{\beta}_{12}) &= \beta_{12} \\ E(\hat{\beta}_{11}) &= \beta_{11} \\ E(\hat{\beta}_{22}) &= \beta_{22} \end{aligned}$$

Example 2

This procedure is very useful when the design is a mixed-level fractional factorial. For example, consider the mixed-level design in Table 9.10 of the textbook. This design can accommodate four two-level factors and a single three-level factor. The resulting resolution III fractional factorial is shown in Table 3.

Since the design is resolution III, the appropriate model contains the main effects

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_{55}(x_5^2 - \bar{x}_5^2) + \varepsilon,$$

where the model terms

$$\beta_5 x_5 \text{ and } \beta_{55}(x_5^2 - \bar{x}_5^2)$$

represent the linear and quadratic effects of the three-level factor x_5 . The quadratic effect of x_5 is defined so that it will be orthogonal to the intercept term in the model.

Table 3. A Mixed-Level Resolution III Fractional Factorial

x ₁	x ₂	x ₃	x ₄	x ₅
-1	1	1	-1	-1
1	-1	-1	1	-1
-1	-1	1	1	0
1	1	-1	-1	0
-1	1	-1	1	0
1	-1	1	-1	0
-1	-1	-1	-1	1
1	1	1	1	1

Now suppose that the **true** model has interaction:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_{55}(x_5^2 - \bar{x}_5^2) \\ + \beta_{12} x_1 x_2 + \beta_{15} x_1 x_5 + \beta_{155} x_1 (x_5^2 - \bar{x}_5^2) + \varepsilon$$

So in the true model the two-level factors x_1 and x_2 interact, and x_1 interacts with both the linear and quadratic effects of the three-level factor x_5 . Straightforward, but tedious application of the procedure described above leads to the alias matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/2 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and the alias relationships are computed from

$$E(\hat{\beta}_1) = \beta_1 + (\mathbf{X}_1' \mathbf{X}_1)^{-1} \mathbf{X}_1' \mathbf{X}_2 \beta_2 \\ = \beta_1 + \mathbf{A} \beta_2$$

This results in

$$\begin{aligned}
E(\hat{\beta}_0) &= \beta_0 \\
E(\hat{\beta}_1) &= \beta_1 \\
E(\hat{\beta}_2) &= \beta_2 + (1/2)\beta_{15} \\
E(\hat{\beta}_3) &= \beta_3 + (1/2)\beta_{15} \\
E(\hat{\beta}_4) &= \beta_4 + (1/2)\beta_{155} \\
E(\hat{\beta}_5) &= \beta_5 + \beta_{12} \\
E(\hat{\beta}_{55}) &= \beta_{55}
\end{aligned}$$

The linear and quadratic components of the interaction between x_1 and x_5 are aliased with the main effects of x_2, x_3 , and x_4 , and the x_1x_2 interaction aliases the linear component of the main effect of x_5 .

S9.3 More about Decomposing Sums of Squares in Three-Level Designs

I am indebted to my colleague Professor Saeed Maghsoodloo of Auburn University who pointed out this interesting approach to decomposing sums of squares and prepared this material for inclusion in the book.

We define an effect in a balanced b^k factorial design (in this chapter $b = 3$) in such a manner that it occupies only one column of the corresponding design matrix and hence it will absorb exactly $(b - 1)$ degrees of freedom (df). This number of df is due to the fact that each column of a b^k factorial design contains b levels 0, 1, 2, ..., $b-1$. As an example, in a 2^3 balanced factorial design, we have 7 effects, A, B, C, AB, AC, BC, ABC, each carrying 1 df and occupying exactly one column, as shown in Table 6.3. While an equispaced 3^2 balanced factorial design has 4 effects, A, B, AB, and AB^2 , each with 2 df , and hence a 3^2 factorial design must have 4 distinct columns that are occupied by the orthogonal effects A, B, AB and AB^2 as shown in Table 9.4 for the data of Example 5.5. This is due to the fact in Table 9.4 the nine FLCs (Factor Level Combinations) provide 8 df for studying orthogonal effects, and each column of the

Table 4. Design Matrix Corresponding to Table 9.4

		AB =J(AB) AB ² = I(AB)		FLC	y _{ijk}
A	B	A + B	A + 2B		
0	0	0	0	00	−2, −1
0	1	1	2	01	−3, 0
0	2	2	1	02	2, 3
1	0	1	1	10	0, 2
1	1	2	0	11	1, 3

1	2	0	2	12	4, 6
2	0	2	2	20	-1, 0
2	1	0	1	21	5, 6
2	2	1	0	22	0, -1

design matrix of Table 9.4 contains 3 levels (0, 1, & 2) and hence must absorb 2 *df*. Because of the exponent 2, in 3^2 , the first 2 columns of Table 9.4 are written completely arbitrarily, but column 3 is obtained from the sum of the first 2 columns modulus 3, while column 4 is obtained from the sum of column 1 and twice column 2 modulus 3. Further, the decomposition of the $A \times B$ interaction in Table 9.4 is valid only if the factors are qualitative, or equispaced quantitative. If levels of a 3-level factor is quantitative but with different spacing, and no transformation can modify the levels into equispacing levels, then the $A \times B$ cannot be decomposed orthogonally into AB and AB^2 . On the other hand, only in base-2 designs, -1 can denote the low level of a quantitative factor and +1 its high level; this is due to the fact that in a 2^k factorial design only linear impacts of quantitative factors on the response can be studied, and the linear contrast coefficients are -1 and +1. The base-2 notation for low and high levels are 0 and 1, respectively. If a factor in a base-2 design is qualitative, it is arbitrary to designate which level as zero and which as 1. In base-3 designs, both the linear and quadratic impacts of quantitative factors on the response can be studied, and for balanced base-3 designs the linear contrast for a quantitative factor is $L_3 = [-1 \quad 0 \quad 1]^T$ while the quadratic contrast is $Q_3 = [1 \quad -2 \quad 1]^T$. Note that the two 3×1 vectors L_3 and Q_3 are orthogonal (or perpendicular) because their dot-product is zero.

In base-3 designs, AB and AB^2 represent the two orthogonal components of the first-order interaction $A \times B$ which has $2 \times 2 = 4$ *df*. In base-2 designs the $A \times B$ interaction has only 1 *df* which can be designated as AB , while for a base-3 design $A \times B$ has 4 *df* and the AB interaction with 2 *df* represents only one of the 2 orthogonal components. Clearly, in base-3 designs $A \times B$ has 4 *df*, and thus it can be embedded into 2 columns of the design matrix 9.4, each of which has 2 *df*. This pattern can be generalized to any prime-bases 5, 7, 11, For example, in a base-5 designs the $A \times B$ interaction has 16 *df*, and if all factors are either qualitative or equispaced quantitative, $A \times B$ can be decomposed into AB , AB^2 , AB^3 , AB^4 each with 4 *df*, and then $A \times B$ can be imbedded into 4 columns of the design matrix.

In order to obtain the AB and AB^2 components of $A \times B$ in 3^2 design, one must use modulus 3 algebra and their contrast functions to obtain their SS's as shown in Table 9.4. While for base-5, one must use modulus 5 algebra to decompose $A \times B$ into AB , AB^2 , AB^3 , AB^4 each with 4 *df*. It is paramount to bear in mind that if the design base is not a prime number, such as 4, then $A \times B$ in base-4 with 9 *df* does not break-down orthogonally into AB , AB^2 , AB^3 components each with 3 *df*. In other words, the effects AB , AB^2 and AB^3

(or AB , AB^3 and A^2B^3) are meaningless in base 4 because they do not form an orthogonal (i.e., additive) decomposition of $A \times B$ interaction (with 9 df). This is due to the fact that there does not exist a transformation that will make the $(\mathbf{X}'\mathbf{X})$ matrix diagonal. Therefore, direct confounding in blocks and direct fractionalizing in bases 4 & 6 is not possible, and one has to resort to pseudo factors.

In general, the b^k balanced factorial design, where b is a prime number, has

$$\sum_{j=0}^{k-1} b^j = \frac{b^k - 1}{b - 1}$$

orthogonal effects (or columns) each with $(b - 1) df$, and only k columns of the main factors in the design matrix can be written arbitrarily, and the rest must be obtained from the use of their contrast functions and the modulus b algebra. For example, a 2^5 factorial has $(2^5 - 1)/(2 - 1) = 31$ effects A, B, ...E, AB, ..., DE, ABC, ..., CDE, ABCD, ..., BCDE, ABCDE each with $1 df$, where the columns for $k = 5$ main factors A, B, C, D & E are written arbitrarily. The 3^3 factorial design has $(3^3 - 1)/(3 - 1) = 13$ orthogonal effects which are A, B, C, AB, AB^2 , AC, AC^2 , BC, BC^2 , ABC, AB^2C , ABC^2 , AB^2C^2 each with $b - 1 = 2 df$, and only the 3 columns A, B, and C are written arbitrarily. Thus, the 3^3 factorial will have $27 = 3^3$ distinct FLCs but only $(3^3 - 1)/(3 - 1) = 13$ distinct effects (or orthogonal columns) each with $2 df$. Similarly, the 5^2 factorial has $(5^2 - 1)/(5 - 1) = 6$ orthogonal effects A, B, AB, AB^2 , AB^3 , AB^4 each with $4 df$. The balanced 5^2 design has 25 distinct FLCs 00, 01, 02, 03, 04, 10, 11, 12, 13, 14, ... 34, 40, ..., 44 that provide $24 df$ for studying model effects and hence its design matrix has 6 orthogonal columns, the first 2 of which for A & B are written arbitrarily, and the 4 non-arbitrary columns are occupied by the $4-df$ effects AB, AB^2 , AB^3 , AB^4 .

The contrast function in base-2 for the effect AB is $\xi(\text{AB}) = x_1 + x_2$, where x_1 represents the levels of factor A (0 for low or 1 for high) and x_2 represents the levels of factor B (0 or 1); the contrast function for the effect AB^2C in a base-3 design is $\xi(\text{AB}^2\text{C}) = x_1 + 2x_2 + x_3$, where x_3 represents the levels of factor C (0 for low, 1 for medium, and 2 for the high level); the contrast function for the effect AB^3 in a base-5 design is $\xi(\text{AB}^3) = x_1 + 3x_2$ (where x_1 and $x_2 = 0, 1, 2, 3, \text{ or } 4$). Note that a contrast function in base 2 can take on only the values of 0, or 1; a contrast function in base 3 can have only the values 0, 1, or 2, while in base 5 a contrast function can have only the mod-5 values 0, 1, 2, 3, or 4.

The design matrix for the data of Table 9.1 is given in Table 5, which shows that the first 3 columns are written arbitrarily, and the other 10 non-arbitrary columns are obtained from the contrast function $\xi = a_1x_1 + a_2x_2 + a_3x_3$, where a_i 's can take on only the

Table 5. The Design Matrix for the 3^3 Factorial of Example 9.1

[illegible]

0	0	1	0	0	1	2	1	2	1	1	2	2	185
0	0	2	0	0	2	1	2	1	2	2	1	1	9
0	1	0	1	2	0	0	1	1	1	2	1	2	-105
0	1	1	1	2	1	2	2	0	2	0	0	1	20
0	1	2	1	2	2	1	0	2	0	1	2	0	-70
0	2	0	2	1	0	0	2	2	2	1	2	1	-25
0	2	1	2	1	1	2	0	1	0	2	1	0	134
0	2	2	2	1	2	1	1	0	1	0	0	2	67
1	0	0	1	1	1	1	0	0	1	1	1	1	41
1	0	1	1	1	2	0	1	2	2	2	0	0	175
1	0	2	1	1	0	2	2	1	0	0	2	2	-28
1	1	0	2	0	1	1	1	1	2	0	2	0	-123
1	1	1	2	0	2	0	2	0	0	1	1	2	-99
1	1	2	2	0	0	2	0	2	1	2	0	1	-126
1	2	0	0	2	1	1	2	2	0	2	0	2	24
1	2	1	0	2	2	0	0	1	1	0	2	1	154
1	2	2	0	2	0	2	1	0	2	1	1	0	-51
2	0	0	2	2	2	2	0	0	2	2	2	2	-74
2	0	1	2	2	0	1	1	2	0	0	1	1	203
2	0	2	2	2	1	0	2	1	1	1	0	0	-85
2	1	0	0	1	2	2	1	1	0	1	0	1	-122
2	1	1	0	1	0	1	2	0	1	2	2	0	-54
2	1	2	0	1	1	0	0	2	2	0	1	2	-113
2	2	0	1	0	2	2	2	2	1	0	1	0	-15
2	2	1	1	0	0	1	0	1	2	1	0	2	245
2	2	2	1	0	1	0	1	0	0	2	2	1	58

y.... = 165

values of mod-3 elements 0, 1, and 2. Further, it is conventional that the first coefficient in the contrast function can be only 0, or 1. For example, the contrast function for column 12 is $\xi(ABC^2) = x_1 + x_2 + 2x_3$ so that for row 20 of this column $\xi(ABC^2) = 2 + 0 +$

$2 \times 1 = 1 \pmod{3}$. All 10 non-arbitrary columns of Table 5 have been obtained in this fashion using modulus-3 algebra and the corresponding column contrast functions.

In order to compute the AC^2 component of $SS(A \times C)$, we have to use column 7 of Table 5 in order to obtain the values of $(AC^2)_0$, $(AC^2)_1$ and $(AC^2)_2$. Table 5 shows that $(AC^2)_0 = -60 - 105 - 25 + 175 - 99 + 154 - 85 - 113 + 58 = -100$; $(AC^2)_1 = 9 - 70 + 67 + 41 - 123 + 24 + 203 - 54 + 245 = 342$; $(AC^2)_2 = 185 + 20 + 134 - 28 - 126 - 51 - 74 - 122 - 15 = -77$.

Therefore, $SS(AC^2) = \frac{(-100)^2 + 342^2 + (-77)^2}{18} - \frac{165^2}{54} = 6878.777\bar{7}$. Similarly, one can verify using column 6 of Table 9.5 that $(AC)_0 = -1$, $(AC)_1 = 141$, and $(AC)_2 = 25$,

which yield $SS(AC) = \frac{(-1)^2 + 141^2 + 25^2}{18} - \frac{165^2}{54} = 635.111\bar{1}$. In order to verify if these 2 components of $SS(A \times C)$ are indeed orthogonal, we will cross factor A with C in order to compute this 4-*df* interaction SS, as depicted in Table 6.

Table 6 The $A \times C$ Interaction

A \ C	10 psi	20	30
Type 1	-190	339	6
Type 2	-58	230	-205
Type 3	-211	394	-140

Table 9.6 shows that $SS(A \times C) =$

$$\frac{(-190)^2 + 339^2 + 6^2 + (-58)^2 + 230^2 + (-205)^2 + (-211)^2 + 394^2 + (-140)^2}{6} -$$

$\frac{165^2}{54} - SS(A) - SS(B) = 7513.888\bar{8}$, where $SS(A) = 993.777\bar{7}$ and $SS(B) = 69105.333\bar{3}$. However, this last $SS(A \times C) = 7513.888\bar{8}$ is exactly equal to $SS(AC) + SS(AC^2) = 635.111\bar{1} + 6878.777\bar{7}$, which confirms the orthogonal breakdown of $A \times C$ into AC and AC^2 components.

In similar fashion, we proceed to compute the $SS(ABC^2)$ using column 12 of Table 5, which shows $(ABC^2)_0 = -60 + 20 + 67 + 175 - 126 + 24 - 85 - 122 + 245 = 138$; $(ABC^2)_1 = 9 - 105 + 134 + 41 - 99 - 51 + 203 - 113 - 15 = 4$; $(ABC^2)_2 = 185 - 70 - 25 - 28 - 123 + 154$

$$-74 - 54 + 58 = 23. \text{ As a result, } SS(ABC^2) = \frac{138^2 + 4^2 + 23^2}{18} - \frac{165^2}{54} = 584.111\bar{1}.$$

Proceeding as above, we obtain $SS(ABC) = 18.777\bar{7}$, $SS(AB^2C) = 221.777\bar{7}$, $SS(AB^2C^2) = 3804.111\bar{1}$. The sum of these last 4 components of $A \times B \times C$ is equal to $18.777\bar{7} + 221.777\bar{7} + 584.111\bar{1} + 3804.111\bar{1} = 4628.777\bar{7}$. However, the SS of the 2nd-order interaction $A \times B \times C$ by definition for this balanced factorial is given by

$$\begin{aligned} SS(A \times B \times C) &= SS_{\text{Model}} - SS_A - SS_B - SS_C - SS(A \times B) - SS(A \times C) - SS(B \times C) \\ &= 162,587.333\bar{3} - 993.777\bar{7} - 61190.333\bar{3} - 69105.333\bar{3} - \\ &\quad 6300.888\bar{8} - 7513.888\bar{8} - 12854.333\bar{3} = 4628.777\bar{7}, \end{aligned}$$

which exactly matches $SS(ABC) + SS(AB^2C) + SS(ABC^2) + SS(AB^2C^2)$. Hence, ABC , AB^2C , ABC^2 , and AB^2C^2 are the 4 orthogonal components of $A \times B \times C$. The complete ANOVA is given in Table 9.2.